

GUI & UI Design

Diogo Cabral
diogo.cabral@isel.pt

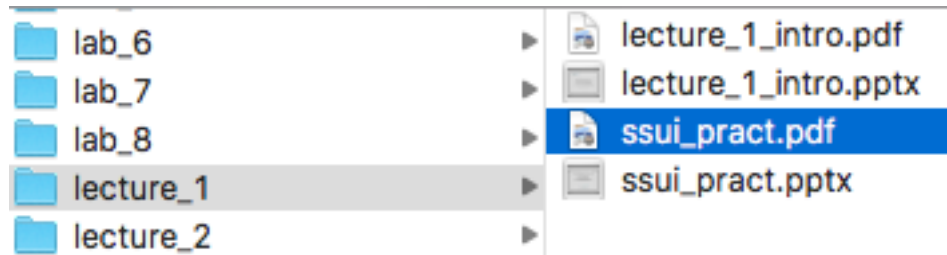
Direct manipulation

Direct manipulation of the object of interest, have the following characteristics:

- Visibility of the object of interest.

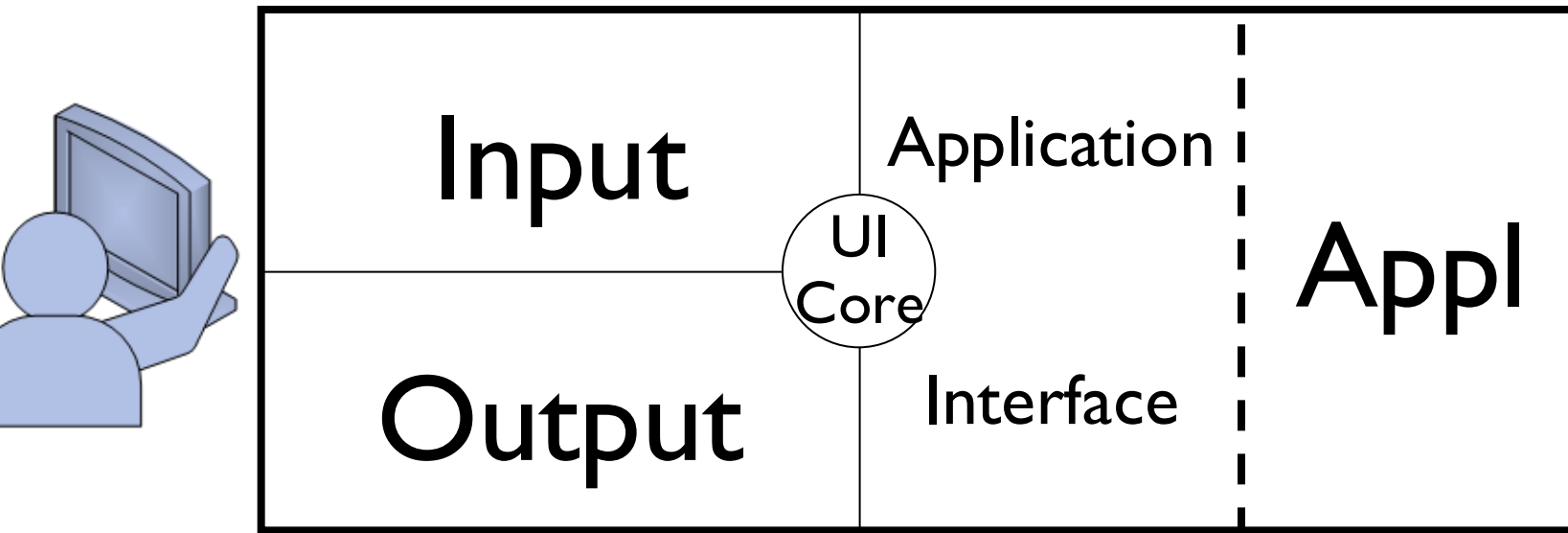
- Rapid, reversible, incremental actions.

- Replacement of complex command language syntax by direct manipulation of the object of interest.



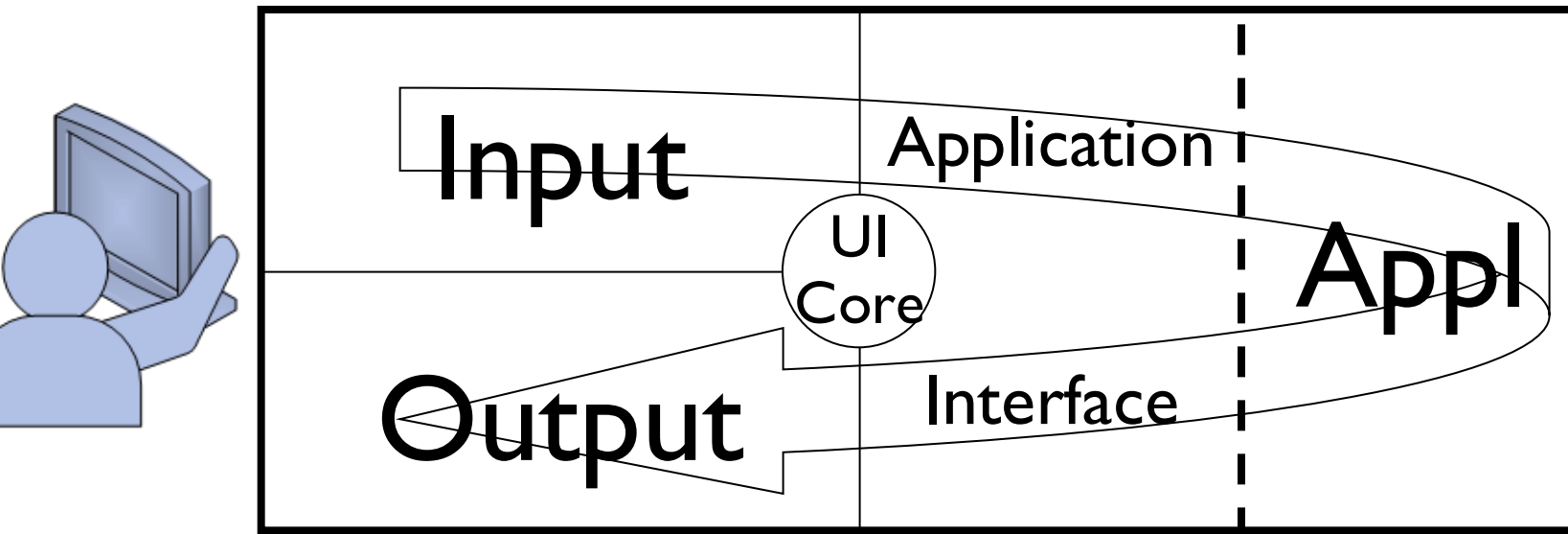
UI tasks

Basic tasks of a UI



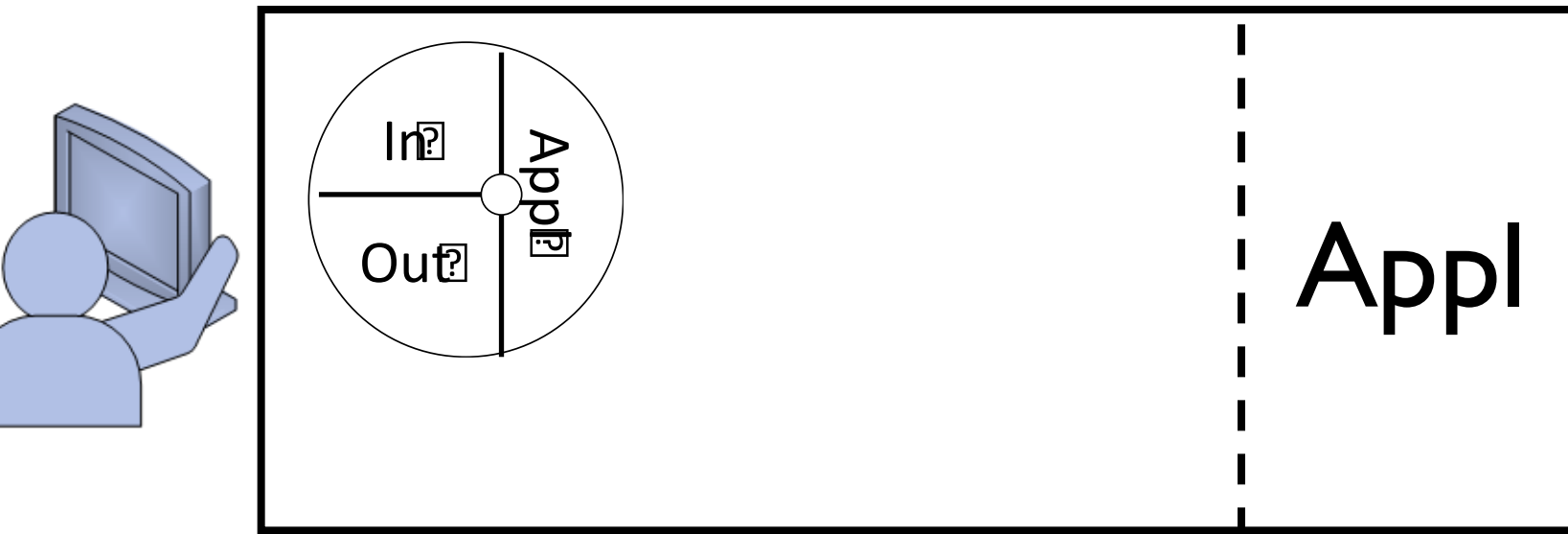
UI tasks

Basic flow



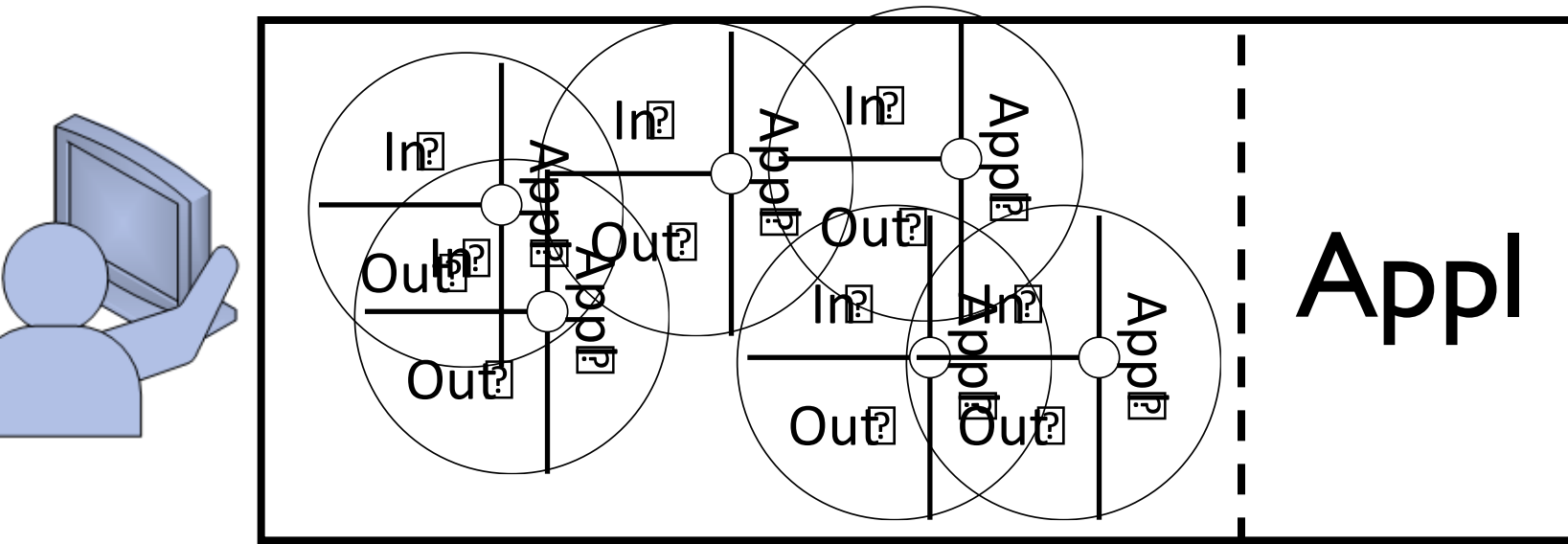
Modern GUIs ("direct manipulation")

Many quasi-independent agents



Modern GUIs ("direct manipulation")

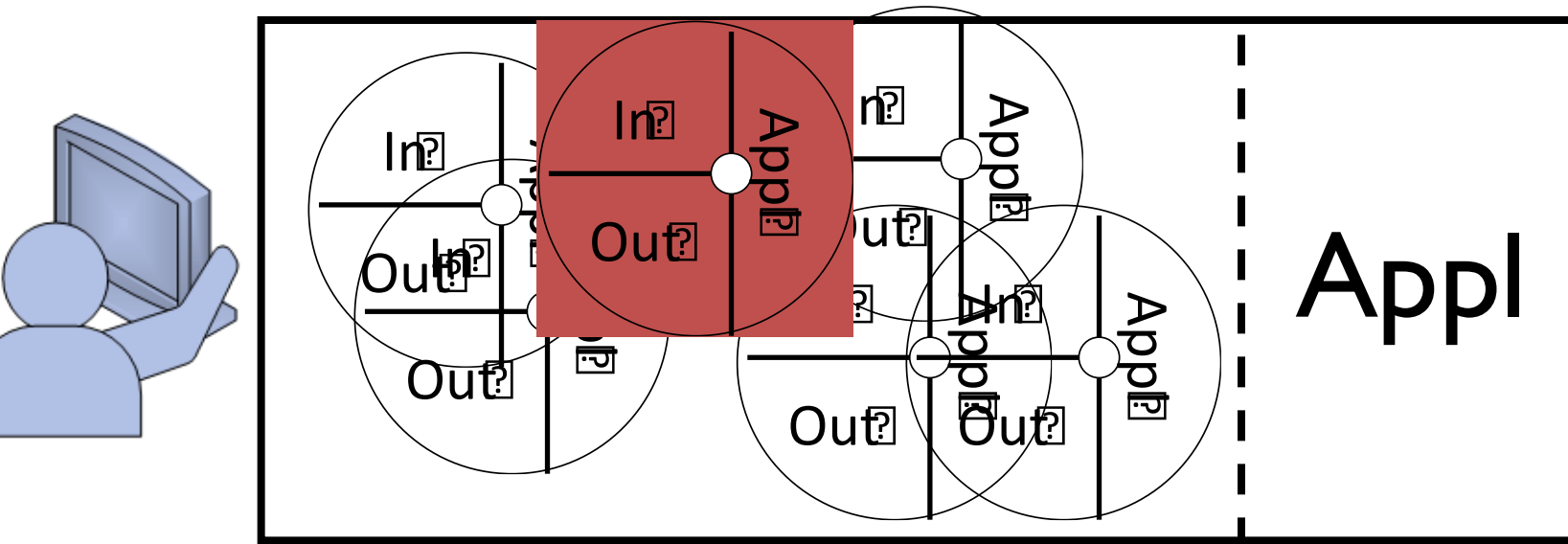
Many quasi-independent agents



Modern GUIs ("direct manipulation")

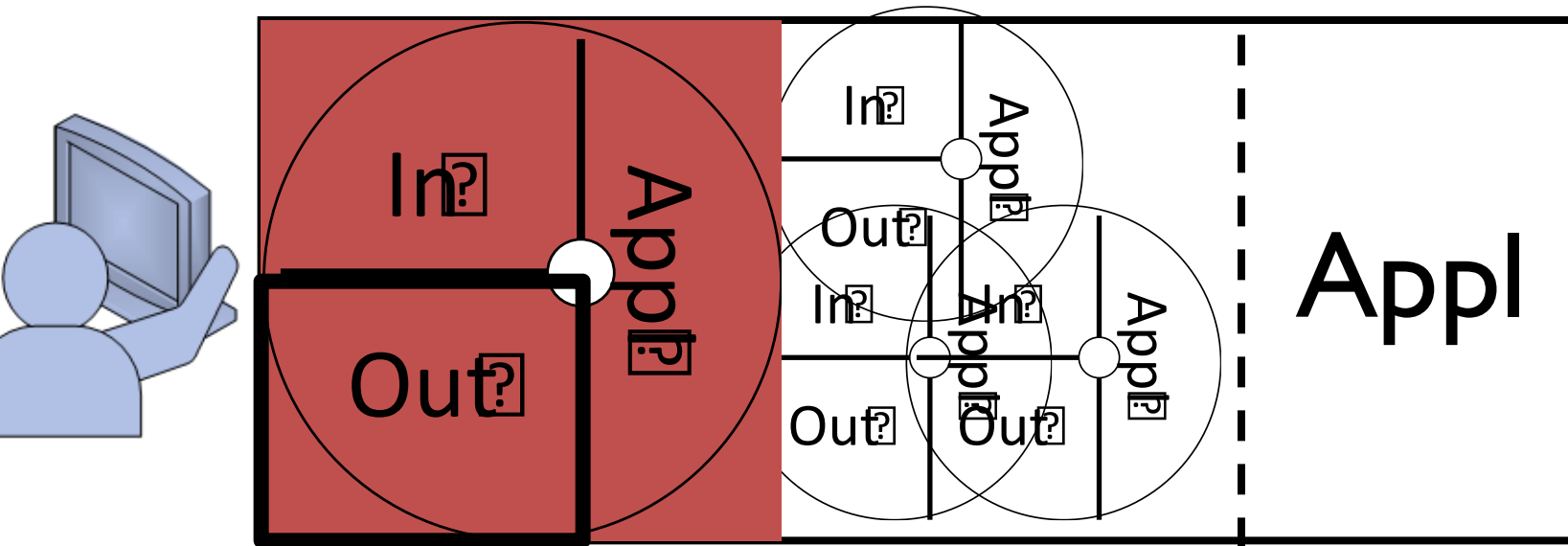
Many quasi-independent agents

Button



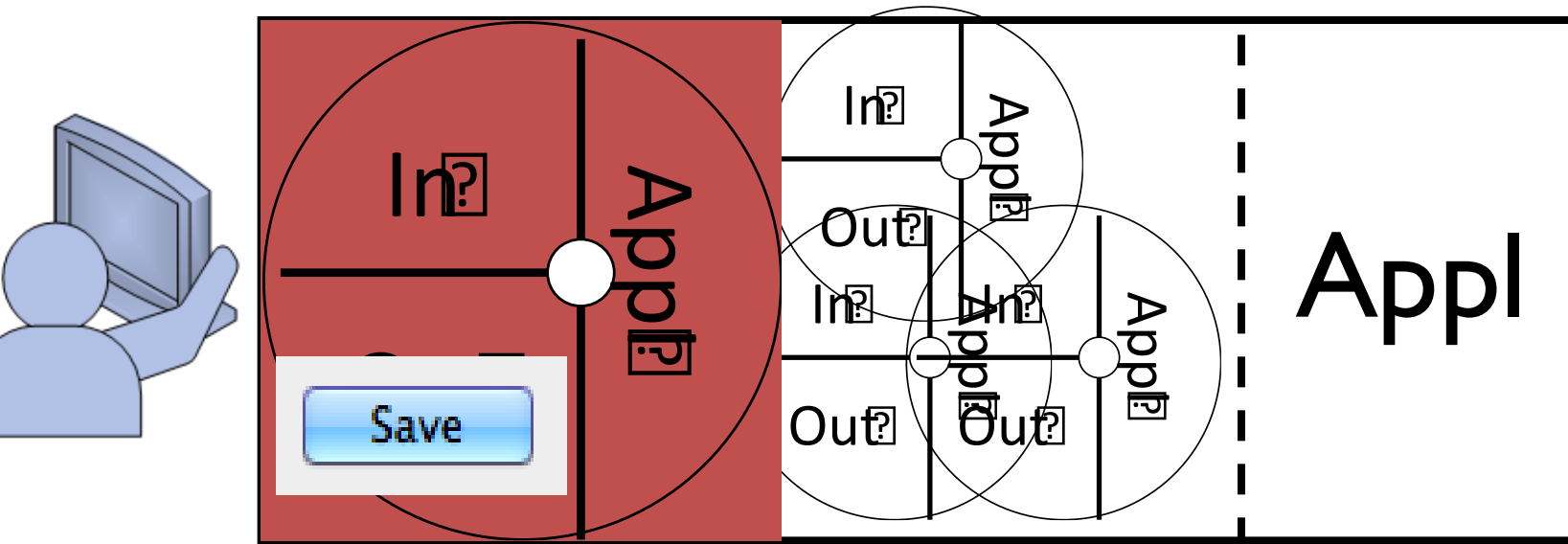
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Modern GUIs ("direct manipulation")

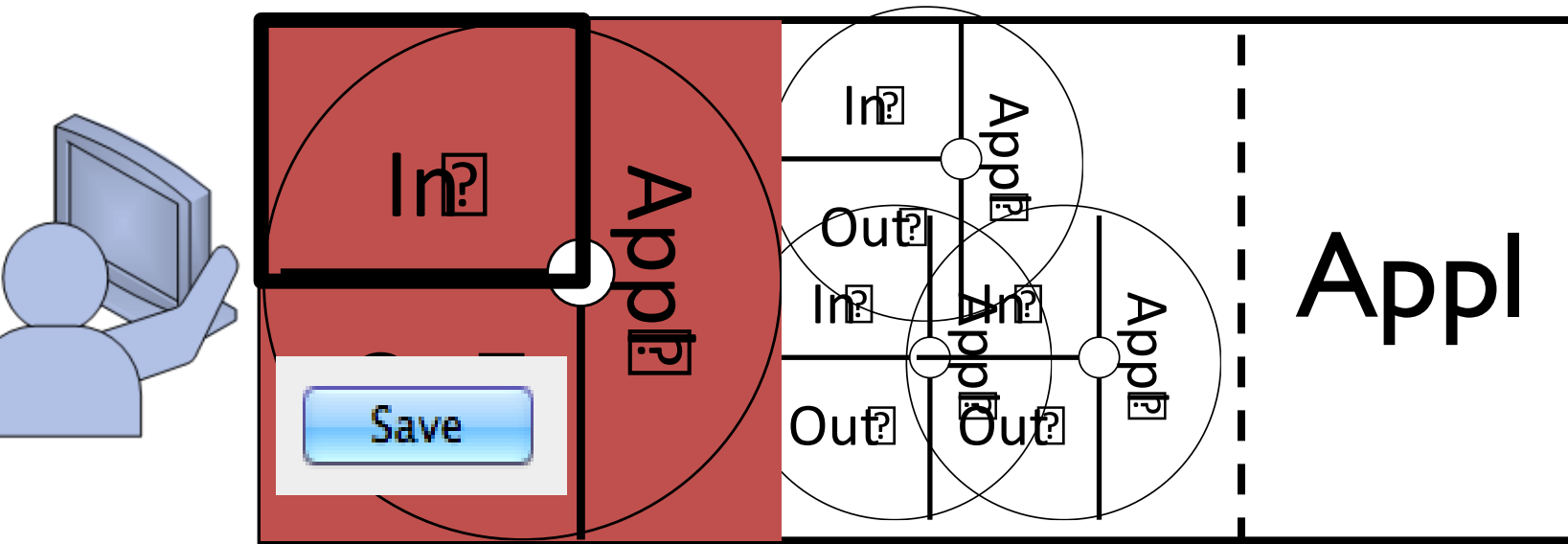
Many quasi-independent agents



Modern GUIs

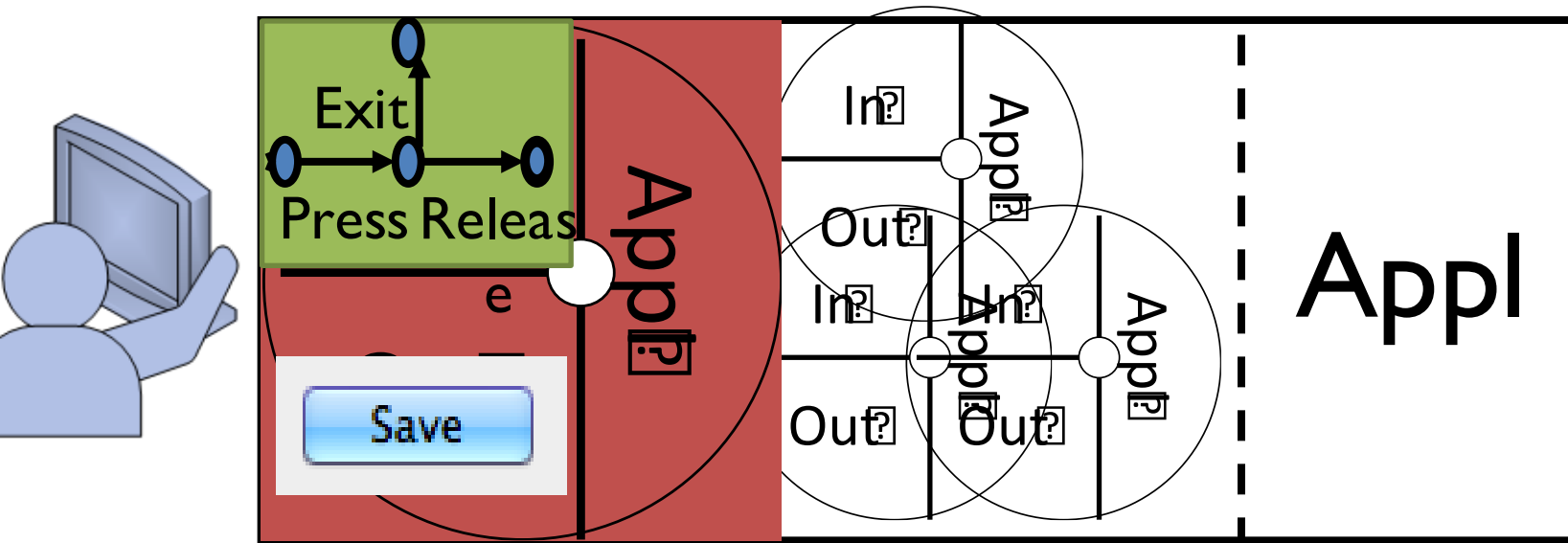
(“direct manipulation”)

Many quasi-independent agents



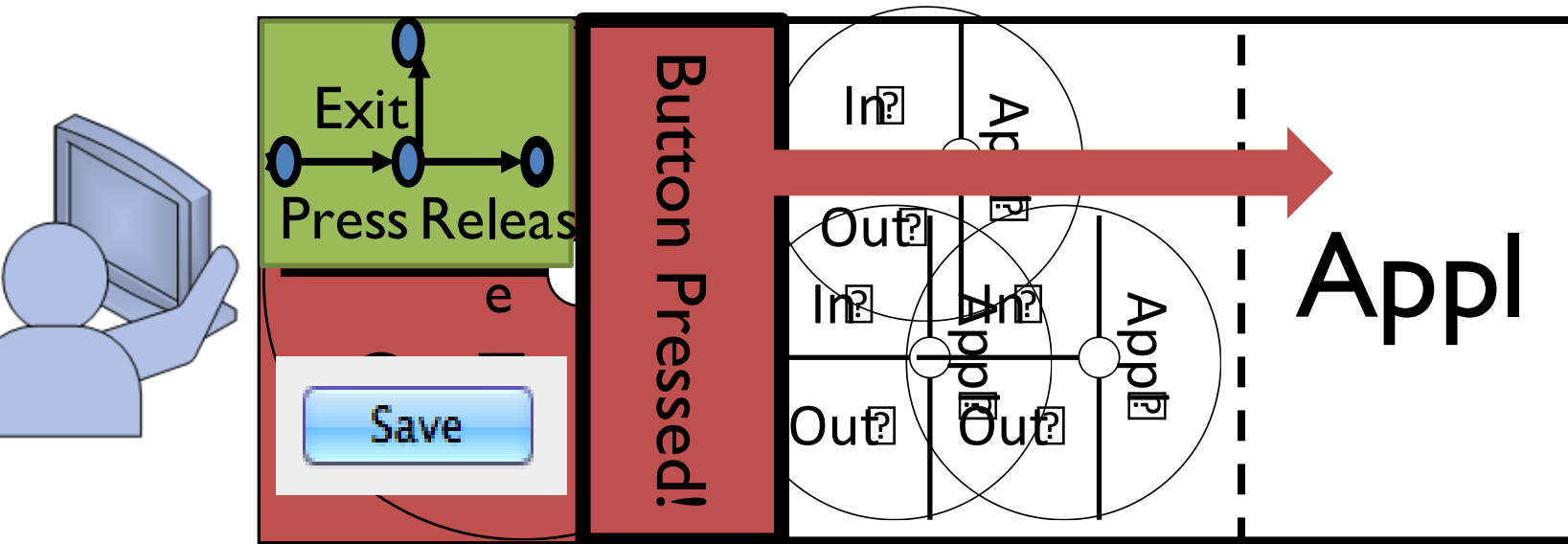
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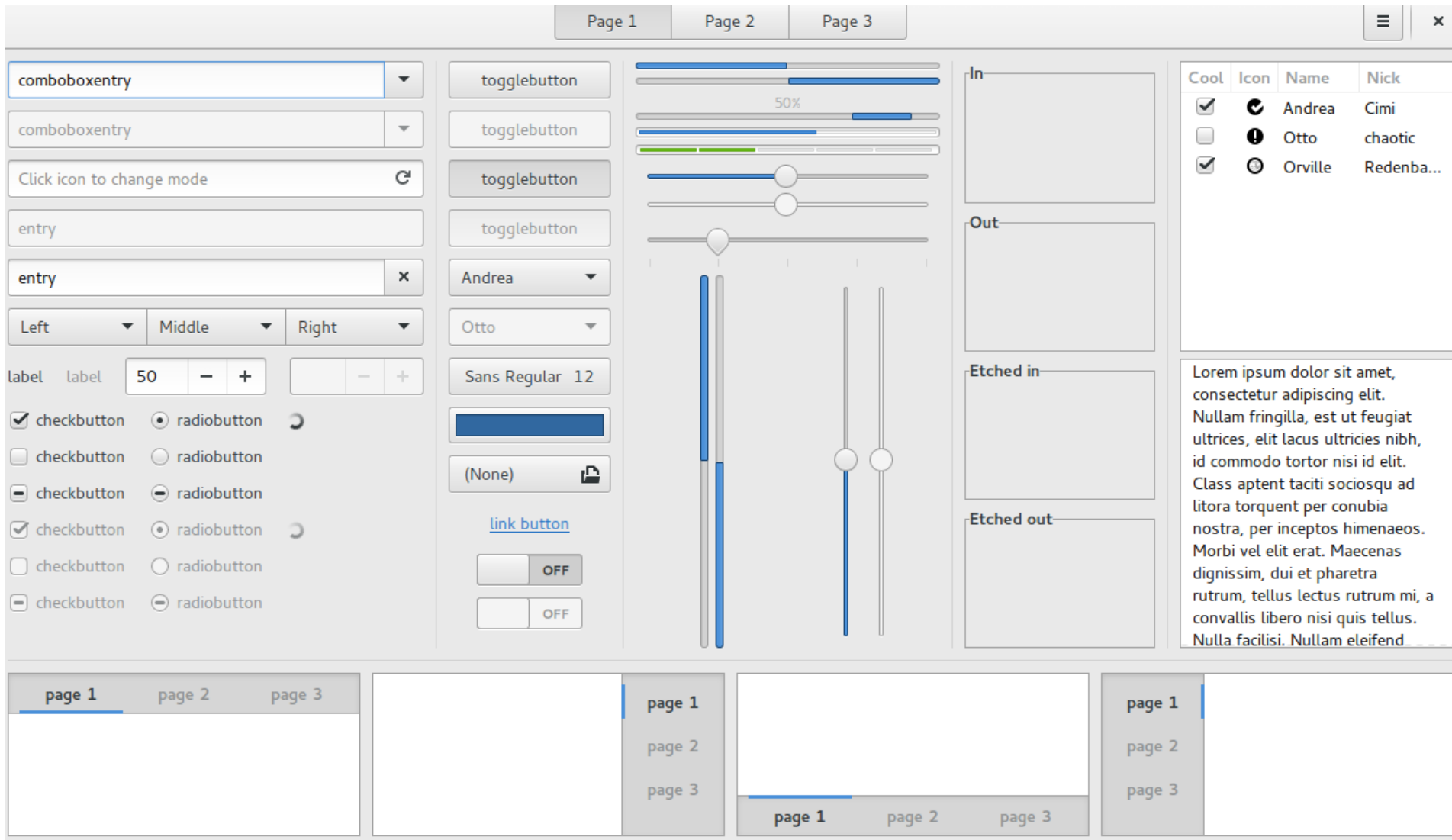


Modern GUIs ("direct manipulation")

Many quasi-independent agents



GUI Widgets



GUI Toolkits

A toolkit is a software library that defines commonly used widgets like menus, buttons, scrollbars, and so on. Examples:

GTK (linux):

https://docs.gtk.org/gtk4/visual_index.html

Qt Designer (multi-platform, C++):

<https://build-system.fman.io/qt-designer-download>

Android Studio (Mobile devices):

<https://developer.android.com/studio/write/layout-editor>

Unity (Games and other):

<https://docs.unity3d.com/6000.2/Documentation/Manual/ui-systems/introduction-ui-toolkit.html>

(.Net MAUI uses XAML.... Not a real toolkit)

Interaction Techniques

Interaction Technique: a computation that couples input and output, to support elementary interactive tasks.

Interaction techniques:

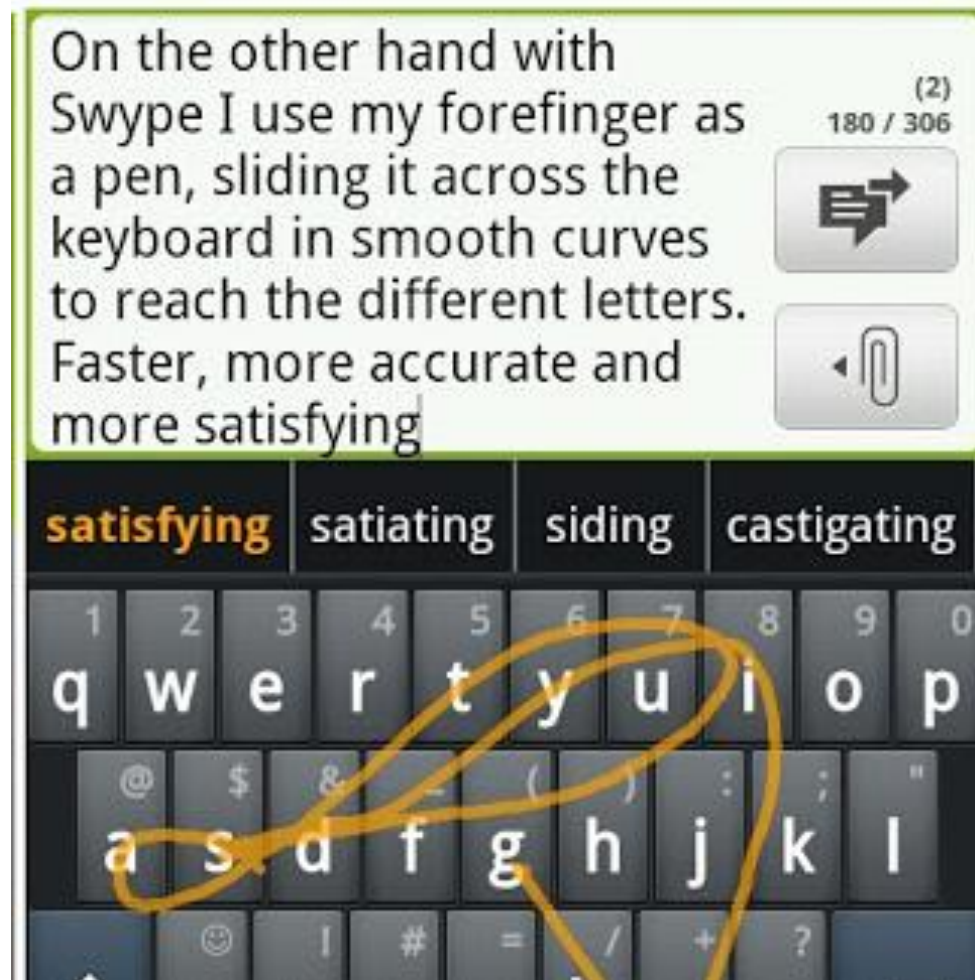
- 1) improve user performance in some conditions of interest.
- 2) enable interaction in circumstances in which it would normally be challenging or even impossible.

If you want to know more about it:

Brad A. Myers. 2024. Pick, Click, Flick! The Story of Interaction Techniques (1st. ed.).
Association for Computing Machinery, New York, NY, USA.

Interaction technique: keyboard swipe. Over 50 WPM!

Most soft keyboards don't get above 20 WPM!



GUI Design

GUIs exhibit great flexibility in how they can be designed.

GUI – Metaphors

Metaphors may be leveraged to guide interface design. The idea behind the use of metaphors is to design concepts, processes, and actions in a GUI around the user's prior knowledge.

-> *matching users' mental model*



Norman's Design Guidelines

Visibility

Feedback

Constraints

Mapping

Consistency

Affordance

Norman's Design Guidelines

Visibility – The more visible functions are, the more likely users will be able to know what to do next. In contrast, when functions are "out of sight," it makes them more difficult to find and know how to use.

Feedback – Feedback is about sending back information about what action has been done and what has been accomplished, allowing the person to continue with the activity. Various kinds of feedback are available for interaction design-audio, tactile, verbal, and combinations of these.

Norman's Design Guidelines

Constraints – The design concept of constraining refers to determining ways of restricting the kind of user interaction that can take place at a given moment. There are various ways this can be achieved.

Mapping – This refers to the relationship between controls and their effects in the world. Nearly all artifacts need some kind of mapping between controls and effects, whether it is a flashlight, car, power plant, or cockpit. An example of a good mapping between control and effect is the up and down arrows used to represent the up and down movement of the cursor, respectively, on a computer keyboard.

Norman's Design Guidelines

Consistency – This refers to designing interfaces to have similar operations and use similar elements for achieving similar tasks. In particular, a consistent interface is one that follows rules, such as using the same operation to select all objects. For example, a consistent operation is using the same input action to highlight any graphical object at the interface, such as always clicking the left mouse button. Inconsistent interfaces, on the other hand, allow exceptions to a rule.

Norman's Design Guidelines

Affordance – is a term used to refer to an attribute of an object that allows people to know how to use it. For example, a mouse button invites pushing (in so doing acting clicking) by the way it is physically constrained in its plastic shell. At a very simple level, to afford means "to give a clue" (Norman, 1988). When the affordances of a physical object are perceptually obvious it is easy to know how to interact with it.

Gestalt Design Guidelines

The central principle to the Gestalt theory was neatly summarized by the Gestalt psychologist Kurt Koffka: "**The whole is *other* than the sum of the parts.**"

The human eye and brain perceive a unified shape in a different way to the way they perceive the individual parts of those shapes. Gestalt principles or laws are rules that describe how the human eye perceives visual elements.

“Gestalt” refers to “shape” or “form” in German.

Gestalt Design Guidelines

1



Simplicity

Combining simplicity with creativity can lead to stunning creations.

How to Master Simplicity:

Know how to balance simple shapes with visual stimulation. Give the eye a comfortable form that helps it interpret what it sees.

2



Figure-Ground

People can immediately identify which element is the figure, and which is the ground. Use these two related principles to make the most of the figure-ground effect:

- **Area** - The viewer's mind sees the smallest element as the figure and the larger one as the ground or background.
- **Convexity** - Convex elements are related to figures.

3



Proximity

Elements close to each other are perceived as part of the same group. Common Use

Case: Kerning. Proper kerning helps readers snap up each word.

4



Similarity

Elements that look alike are perceived as part of a group. The principle of similarity applies to:

- Color
- Shape
- Size
- Texture
- Orientation

<https://www.designcontest.com/blog/designing-with-gestalt-principles/>

Gestalt Design Guidelines

5



Common Fate

Objects that seem to be moving in the same direction are often seen as a group.

6



Symmetry

The principle of symmetry applies to

- Mirrored shapes
- Balanced elements
- Parallel lines

7



Continuity

Objects that are plotted in a continuous pattern are grouped together by the mind. Smooth lines often make a unified figure.

8



Closure

The mind wants closure. A shape only needs to be implied for the mind to "fill in the gaps" and see what it wants to see

<https://www.designcontest.com/blog/designing-with-gestalt-principles/>

Gestalt Design Guidelines

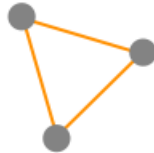
9



Common Region

Elements in one region are seen as one group. Badge designs are seen as unified, despite some containing text, banners, and other objects.

10



Connectedness

When there is a connection between elements, it is easy to see a unified figure.

Achieve connections with:

- Arrow
- Lines
- Illustrations (i.e. a rope)

Gestalt Theory

Humans are bombarded with countless signals day-to-day. To keep from going crazy, we unify these signals into groups. Gestalt designers are obsessed with how people put objects together in their minds.

Good designs leads people to experience the message you want to convey.

Arm yourself with the principles of Gestalt and create designs that have the impact you want.

<https://www.designcontest.com/blog/designing-with-gestalt-principles/>

Gestalt Design

Ungrouped	Proximity	Common area	Similarity	Continuation
Email Word Excel Powerpoint Drive Mastodon Facebook Snap Instagram WhatsApp	Email Word Excel Powerpoint Drive Mastodon Facebook Snap Instagram WhatsApp	Email Word Excel Powerpoint Drive Mastodon Facebook Snap Instagram WhatsApp	Email Word Excel Powerpoint Drive Mastodon Facebook Snap Instagram WhatsApp	Email Word Excel Powerpoint Drive Mastodon Facebook Snap Instagram WhatsApp

Shneiderman's "Eight Golden Rules of Interface Design"

- 1 Strive for consistency.**
- 2 Enable frequent users to use shortcuts.**
- 3 Offer informative feedback.**
- 4 Design dialog to yield closure.**
- 5 Offer simple error handling.**
- 6 Permit easy reversal of actions.**
- 7 Support internal locus of control.**
- 8 Reduce short-term memory load.**

Shneiderman's "Eight Golden Rules of Interface Design"

1 Strive for consistency.

Consistent sequences of actions should be required in similar situations; identical terminology should be used in prompts, menus, and help screens; and consistent commands should be employed throughout.

2 Enable frequent users to use shortcuts.

As the frequency of use increases, so do the user's desires to reduce the number of interactions and to increase the pace of interaction. Abbreviations, function keys, hidden commands, and macro facilities are very helpful to an expert user.

3 Offer informative feedback.

For every operator action, there should be some system feedback. For frequent and minor actions, the response can be modest, while for infrequent and major actions, the response should be more substantial.

Shneiderman's "Eight Golden Rules of Interface Design"

4 Design dialog to yield closure.

Sequences of actions should be organized into groups with a beginning, middle, and end. The informative feedback at the completion of a group of actions gives the operators the satisfaction of accomplishment, a sense of relief, the signal to drop contingency plans and options from their minds, and an indication that the way is clear to prepare for the next group of actions.

5 Offer simple error handling.

As much as possible, design the system so the user cannot make a serious error. If an error is made, the system should be able to detect the error and offer simple, comprehensible mechanisms for handling the error.

6 Permit easy reversal of actions.

This feature relieves anxiety, since the user knows that errors can be undone; it thus encourages exploration of unfamiliar options. The units of reversibility may be a single action, a data entry, or a complete group of actions.

Shneiderman's "Eight Golden Rules of Interface Design"

7 Support internal locus of control.

Experienced operators strongly desire the sense that they are in charge of the system and that the system responds to their actions. Design the system to make users the initiators of actions rather than the responders.

8 Reduce short-term memory load.

The limitation of human information processing in short-term memory requires that displays be kept simple, multiple page displays be consolidated, window-motion frequency be reduced, and sufficient training time be allotted for codes, mnemonics, and sequences of actions.

Mobile App Paper Prototype

<https://www.youtube.com/watch?v=y20E3qBmHpg>

<https://www.nngroup.com/articles/paper-prototyping-cutout-kit/>

<https://medium.com/@karenavongampai/rapid-prototyping-mobile-app-paper-prototype-fa3ea0bc7430>

<https://www.parallelhq.com/blog/what-paper-prototype>